Ocular Adnexa

Images used: 2.2, 2.5, 2.10, 2.1, 2.8, 2.9, 2.12, 2.14

I. External features of the eyelids

A. Palpebral fissure

B. Lateral and medial canthus

1. Plica semilunaris

2. Caruncle

C. Eyelid Margin

1. Cilia

2. Pores of the meibomian glands

II. Anatomy of the eyelids

A. Skin

B. Palpebral conjunctiva

1. Fornix

2. Goblet cells

C. Mucocutaneous junction

D. Tarsal plate

1. Dense connective tissue that gives the eyelid rigidity and structure

2. The superior tarsal plate is approximately 8 mm high in Asians and 10 mm in Caucasians.

3. The inferior tarsal plate is approximately 5 mm high

4. Clinical correlate: Eyelid eversion

E. Orbital septum

1. The orbital septum is a sheet of connective tissue that starts at the orbit rim and inserts into the levator aponeurosis above the tarsal plate superiorly and the tarsal plate inferiorly.

2. Function

    a. Separates the eyelid structures from the orbital contents.

    b. Prevents eyelid infections from entering the orbit and therefore having a direct connection to the brain.

F. Eyelid muscles

1. Orbicularis oculi muscle

    a. Encircles the palpebral fissure.

    b. Functions

        (1) Closes the eyelids

        (2) Protects the eye

        (3) Renews the tear film

        (4) Keeps the eyelid margin near the globe

        (5) Helps with meibomian gland secretion

    c. Clinical correlate: ectropion and entropion

    d. Innervation: cranial nerve 7

2. Superior palpebral levator muscle

    a. The levator muscle starts in the posterior orbit.

    b. It travels forward with the superior rectus muscle.

    c. Levator aponeurosis

        (1) Tendinous expansion of the levator

    d. Function

        (1) Elevation of the eyelids

        (2) Connections between the superior rectus and the levator results in increased eyelid height when looking up.

    e. Innervation: cranial nerve 3

3. Müller muscle

    a. Smooth muscle

    b. Inserts into the tarsal plate

    c. Function

        (1) Additional eyelid elevation

    d. Innervation: sympathetic nervous system

4. Clinical correlate: Ptosis
    a. A drooping of the upper eyelid.

G. Glands of the eyelids

1. Meibomian glands

    a. Embedded in the tarsal plate

    b. Function: secrete the outer lipid layer of the tear film

2. Zeis glands

    a. Secretes sebum into the hair follicle

    b. Function: keep the eyelash from becoming brittle.

3. Accessory lacrimal glands

    a. Located in the upper and lower conjunctival fornix and orbital borders of the tarsal plate

    b. Function: contributes to the aqueous layer of the tear film

III. Lacrimal system

A. Tear film

1. Functions

a. Lubricant between the globe and eyelids

b. Helps remove debris

c. Source of oxygen

d. Antibacterial substances

e. Maintains corneal hydration

f. Regulate ocular surface wound repair

2. Layers

a. Lipid layer

(1) Produced in meibomian glands

(2) Function

i. Retards evaporation

ii. Provides lubrication

iii. Stabilizes the tear film (lowering surface tension) to keep tears from overflowing down the cheeks

b. Aqueous layer

(1) Secreted by the lacrimal and accessory lacrimal glands

c. Mucin layer

(1) The surface of the conjunctival and corneal cells has microvilli covered with this glycocalyx layer which is hydrophilic.

(2) Function

i. Facilitates adhesion of the aqueous layer to the ocular surface

ii. Allows the aqueous layer to spread evenly over the surface of the eye

B. Nasolacrimal drainage system

1. Puncta

2. Canaliculi

3. Lacrimal sac

4. Nasolacrimal duct

5. Valve of Hasner: found at the end of the nasolacrimal duct.